

# MAT 527: Functions of a Complex Variable I

## Submission I

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May 24, 2026

**Theorem** If  $\{U_\lambda\}_{\lambda \in \Lambda}$  is a family of open sets, then  $\bigcup_{\lambda \in \Lambda} U_\lambda$  is open.

**Proof:** Let  $\{U_\lambda\}_{\lambda \in \Lambda}$  be a family of open sets, where  $\Lambda$  is some index set. Let  $z \in \bigcup_{\lambda \in \Lambda} U_\lambda$ . Then for some  $\mu \in \Lambda$ ,  $z$  is an element of  $U_\mu$ . Now, since  $U_\mu$  is open, there exists  $\Delta > 0$  such that  $B(z; \Delta) \subseteq U_\mu$ . But clearly  $U_\mu \subseteq \bigcup_{\lambda \in \Lambda} U_\lambda$ . Therefore,  $B(z; \Delta) \subseteq \bigcup_{\lambda \in \Lambda} U_\lambda$ . Therefore,  $\bigcup_{\lambda \in \Lambda} U_\lambda$  is open. ■